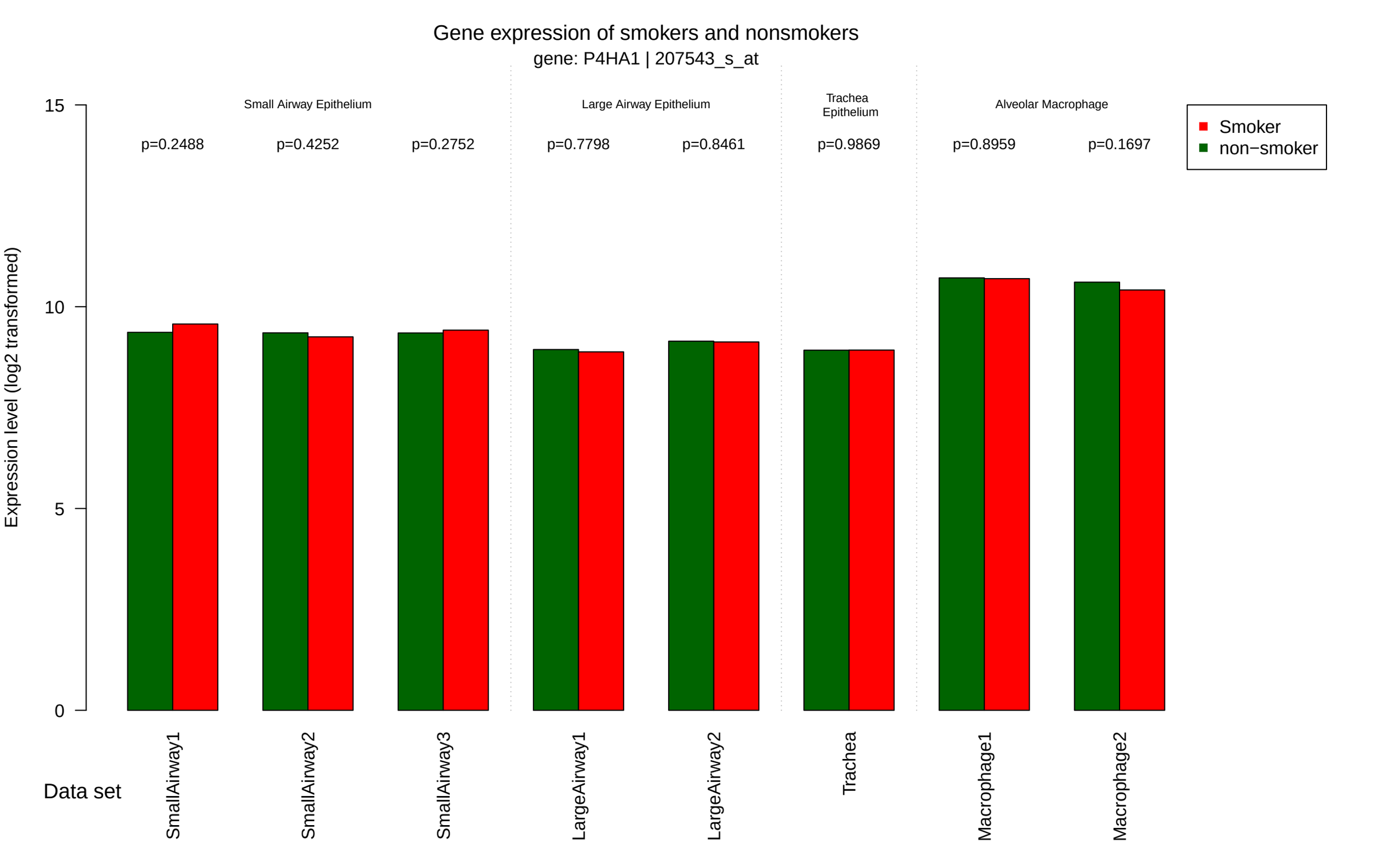
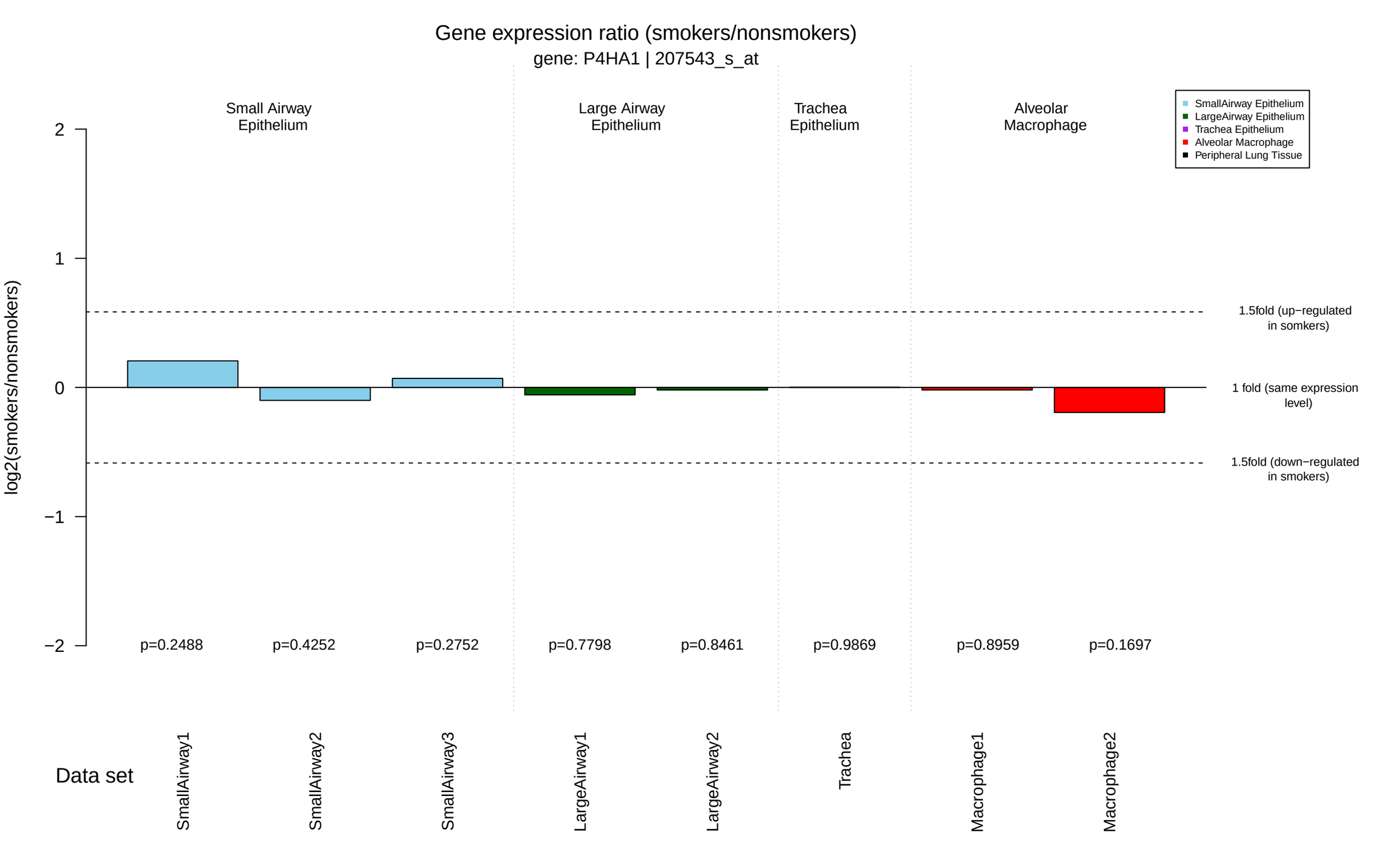
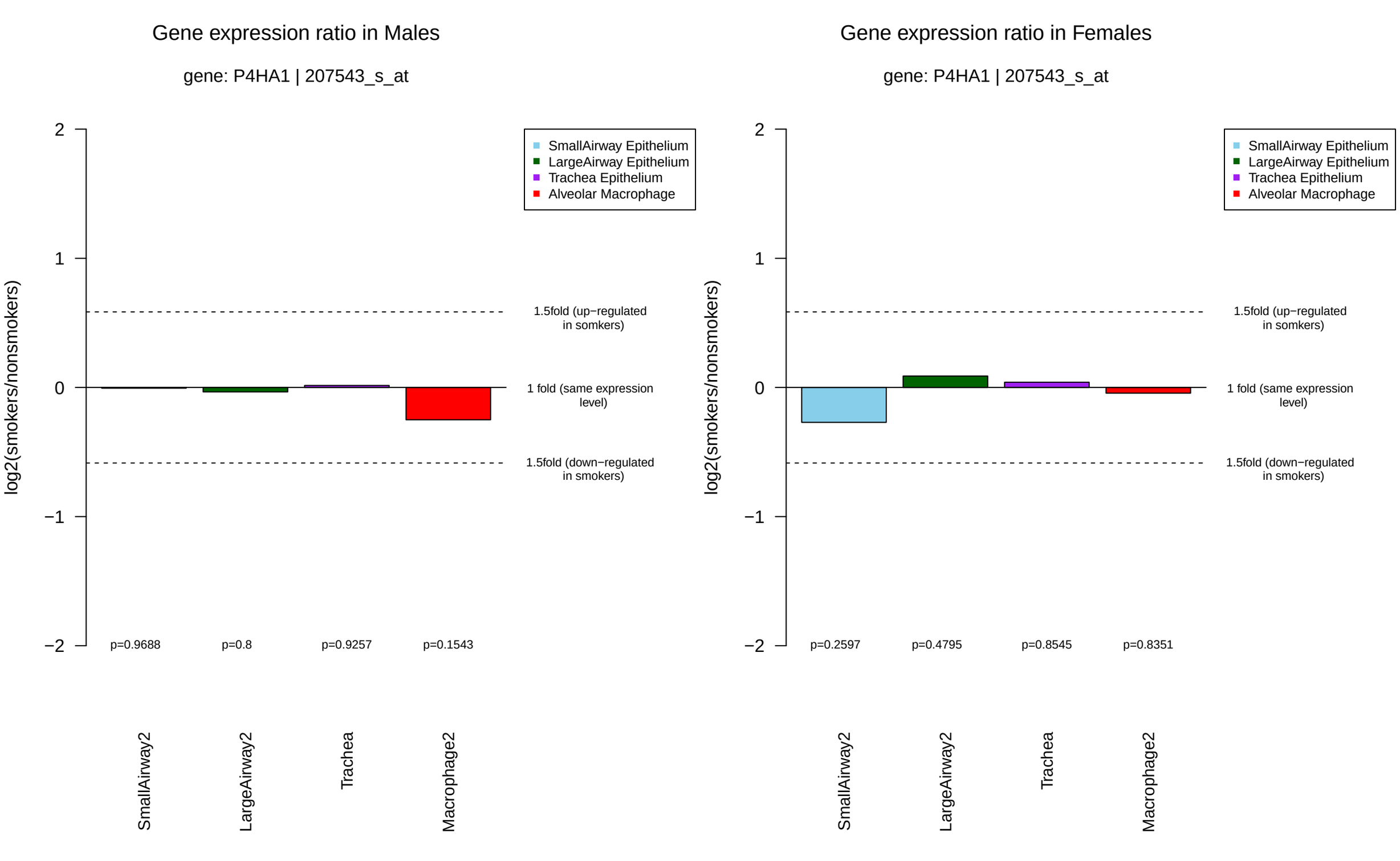
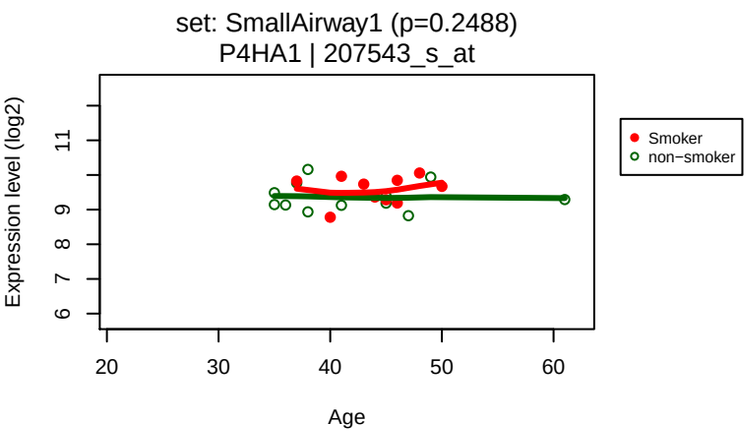
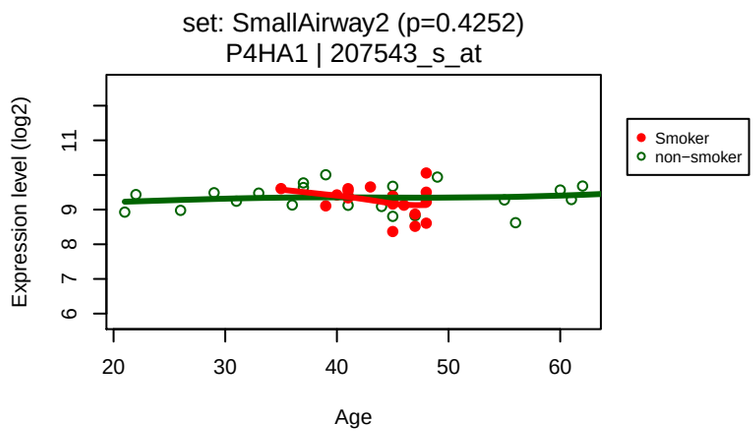
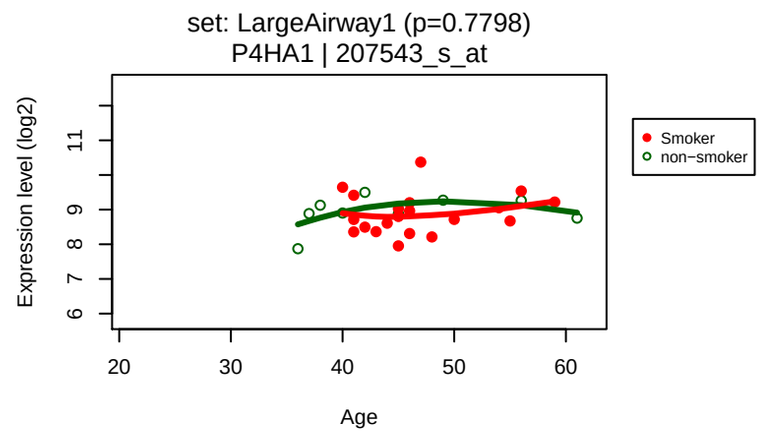

Gene expression of smokers and nonsmokers
gene: P4HA1 | 207543_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2488
p=0.4252
p=0.2752
p=0.7798
p=0.8461
p=0.9869
p=0.8959
p=0.1697
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

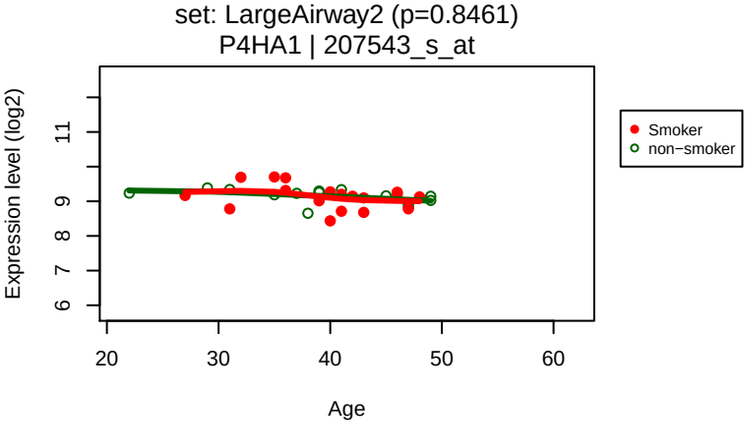

Gene expression ratio (smokers/nonsmokers)
gene: P4HA1 | 207543_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2488
p=0.4252
p=0.2752
p=0.7798
p=0.8461
p=0.9869
p=0.8959
p=0.1697
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

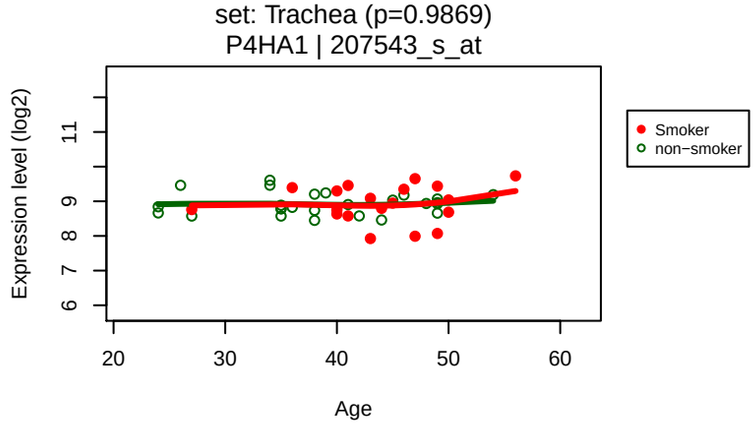

Gene expression ratio in Males
gene: P4HA1 | 207543_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9688
p=0.8
p=0.9257
p=0.1543
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: P4HA1 | 207543_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2597
p=0.4795
p=0.8545
p=0.8351
SmallAirway2
LargeAirway2
Trachea
Macrophage2

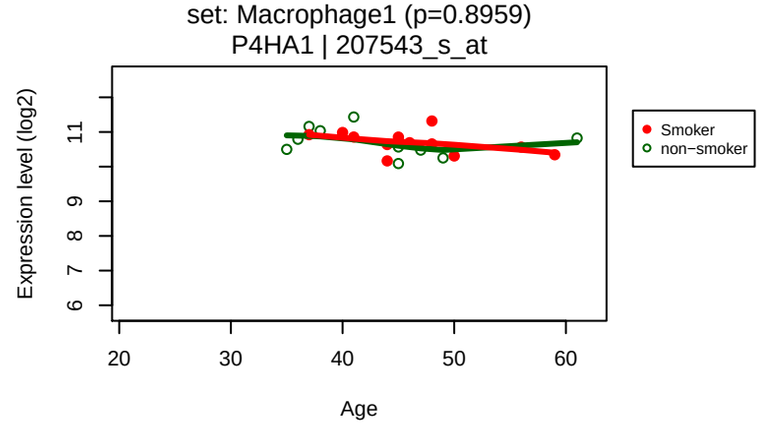

set: SmallAirway1 (p=0.2488)
P4HA1 | 207543_s_at
Expression level (log2)
Age
Smoker
non−smoker

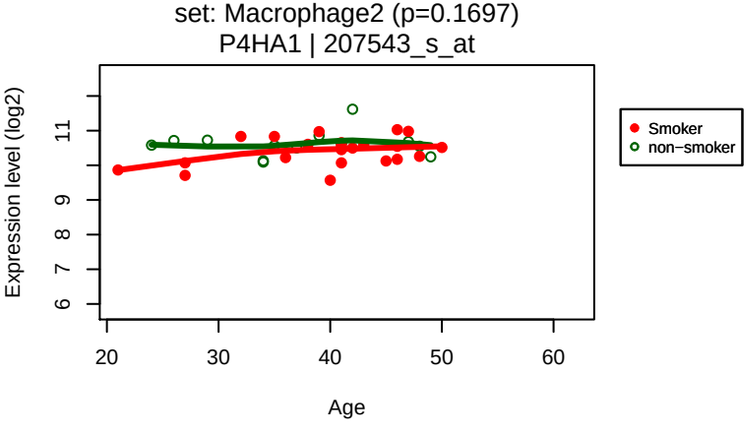

set: SmallAirway2 (p=0.4252)
P4HA1 | 207543_s_at
Expression level (log2)
Age
Smoker
non−smoker

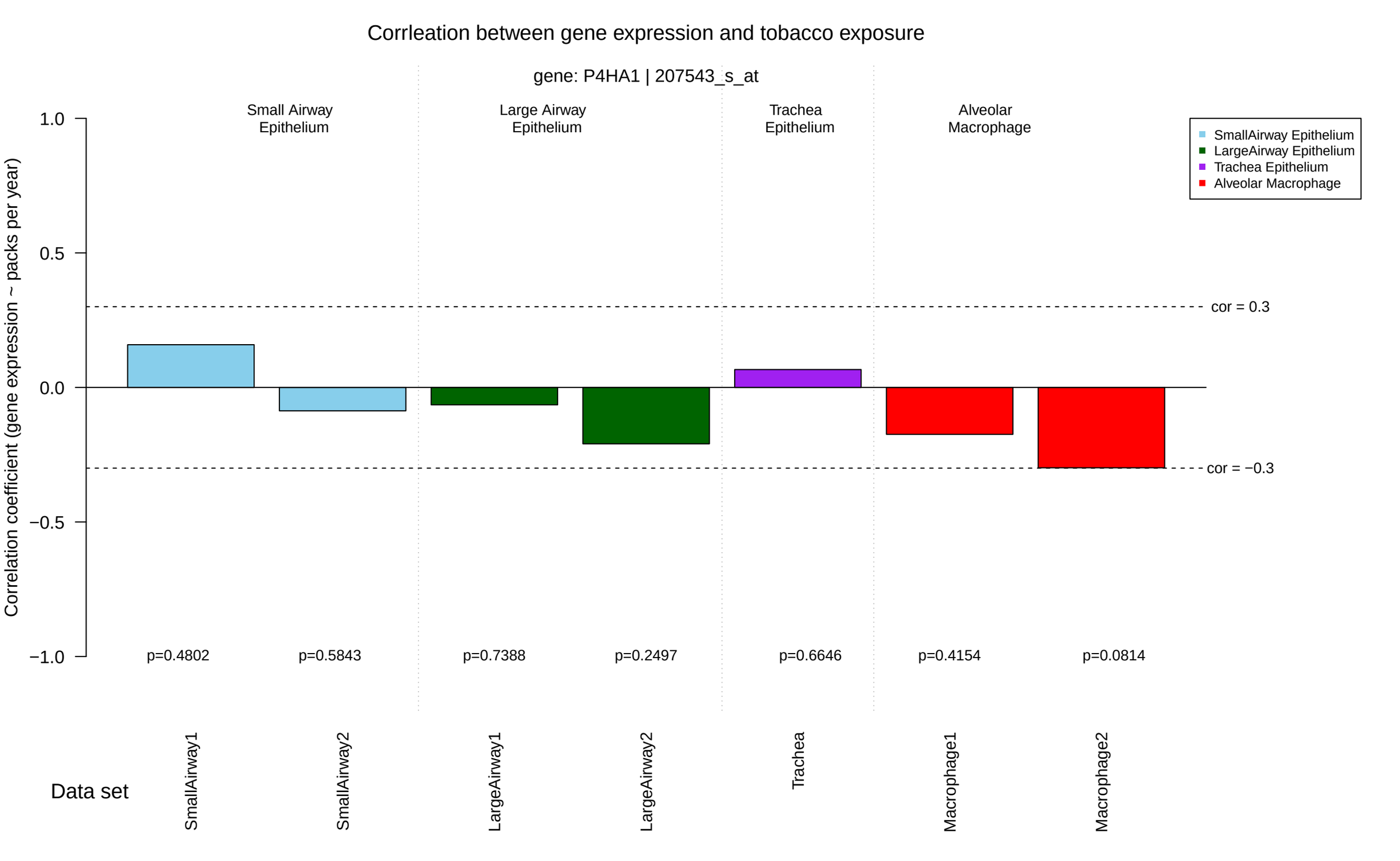

set: LargeAirway1 (p=0.7798)
P4HA1 | 207543_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8461)
P4HA1 | 207543_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9869)
P4HA1 | 207543_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8959)
P4HA1 | 207543_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1697)
P4HA1 | 207543_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: P4HA1 | 207543_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4802
p=0.5843
p=0.7388
p=0.2497
p=0.6646
p=0.4154
p=0.0814
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2